RACHNA KADAM

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EDUCATION

New York University, New York, NY **May 2026**
Master of Science, Mechatronics and Robotics GPA: 3.61
Coursework: *Robotic Kinematics & Dynamics, Advanced Mechatronics, Linear Mathematics, Networked Robotics Systems, Reinforcement Learning, Automated Mobile Robotics*

NMIMS University, Mumbai, India **May 2024**
Bachelor of Technology, Mechatronics GPA: 3.69
Coursework: *Modern Control System, Dynamic System Analysis, Dynamic System Modelling, Fluid Power Automation, Robotic Process Automation, Electric & Hybrid Automotive Technology, PLC & Instrumentation, Human Machine Interface, SCADA*

SKILLS

- **Languages & Compute:** Python (OpenCV, PyTorch, NumPy), C++, C, MATLAB/Simulink, Linux (Ubuntu).
- **Robotics & Middleware:** ROS (Robot Operating System), Kinematics & Dynamics, Localization, Path Planning, Sensor Fusion (IMU, Encoders, Vision).
- **AI & Perception:** Vision Language Models (Moondream), LLM Integration (Gemini,LLaMA), RAG Architecture, Transformers (TF-BERT), Object Detection (ArUco Markers).
- **Control Systems:** Classical Control (PID, LQR), System Identification, State Estimation, Model-Based Design, Real-Time Feedback Loops.
- **Embedded & Hardware:** Raspberry Pi, ESP32, Arduino, Communication Protocols (ESP-NOW, Wi-Fi, BLE, I2C/UART, SPI), PLC (Ladder Logic), SCADA/HMI.
- **Design & Tools:** SolidWorks, Autodesk Fusion 360, Ansys (Simulation), 3D Printing, Git/GitHub, UiPath (RPA).

RESEARCH EXPERIENCE

Mechatronics, Controls and Robotics Lab | Dr. Vikram Kapila **June 2025 - Present**
VLM–LLM Integrated Multi-Robot System for Autonomous Task Execution
- Engineered a "Zero-Code" embodied AI framework integrating Gemini 2.5 Pro and Moondream VLM to decompose natural language into synchronized multi-robot tasks.
- Implemented a hybrid control strategy that combines RRT* for collision-aware planning with a Pure Pursuit + PID switching controller, enabling precise pick-and-place operations in dynamic environments.
- Developed a scalable perception system by applying homography stitching to multiple overhead cameras, creating a unified top-down view for global robot localization and semantic object tracking.
- Orchestrated real-time fleet coordination by leveraging a centralized RPC architecture (ZeroMQ) to synchronize high-level AI logic with low-level robot states across the distributed system.

EXPERIENCE

New York University, Tandon School of Engineering **Sep 2025 – Present**
Course Assistant – Automatic Controls Laboratory
- Assisted in conducting laboratory experiments on system modeling, identification, and real-time control using MATLAB/Simulink and Quanser QuaRC.
- Guided students in implementing PD, PI, PID, and LQR controllers on DC-motor, magnetic levitation, rotary inverted pendulum, and coupled-tank systems.
- Performed hardware validation and troubleshooting on Quanser systems (Maglev, Rotary Pendulum), resolving electromechanical failures and sensor noise issues to ensure 99% up-time for laboratory sessions.
- Designed and deployed a new laboratory module on "Digital Filter Design," teaching students to simulate 2nd-order Butterworth filters in Python, perform Zero-Order Hold (ZOH) discretization, and implement real-time noise suppression on MPU9250 IMUs using Arduino UNO R4.

Lawyantra, Mumbai, India **Dec 2023 – May 2024**
AI Systems Engineer
- Engineered a high-performance NLP pipeline using RAG architecture and TF-BERT, optimizing inference time for large-scale data retrieval.
- Built a bespoke legal search engine with semantic and keyword capabilities, incorporating advanced retrieval algorithms such as TF-BERT, BM25, TF-IDF to improve document ranking.
- Optimized Generative AI models like LLaMA and Mistral for high-precision summarization by fine-tuning parameters, enhancing summary accuracy, and reducing storage overhead using Hugging Face frameworks.

MPSTME Racing Team (ATV) **Sep 2021 – May 2022**
Brakes Department
- Designed, validated, and tested the hydraulic braking circuit for an All-Terrain Vehicle (ATV) using CAD software, including Fusion360 and SolidWorks, achieving precise functionality and robust design.
- Executed comprehensive validation processes utilizing Ansys to confirm simulation results, meeting rigorous performance criteria across 15 testing conditions while enhancing reliability in high stress driving situations during field trials.
- Assembled and tested the braking system seamlessly with the entire vehicle, ensuring reliable performance during competitive trials and compliance with all required safety standards.
- Achieved 10th rank in Phase 2 (Sales) of SAE BAJA, demonstrating exceptional presentation and business strategy skills in promoting the ATV design to industry experts and judges.

PROJECTS

Interactive Motion-Controlled Game Using Computer Vision and IMUs **April 2025– May 2025**
- Designed and implemented a real-time interactive game using OpenCV and MPU6050 IMUs to detect punch direction and intensity via computer vision and sensor fusion.
- Integrated Raspberry Pi, Arduino Nano, and ESP32 for wireless data flow between vision and inertial modules, achieving $<$ 10ms latency.
- Developed the game engine and UI using PyGame, enabling quadrant-based scoring and dynamic feedback.
- Overcame hardware bottlenecks through optimized processing; laid groundwork for multiplayer and fitness-tracking extensions.

Indoor Mapping Robot with Custom SONAR and Real-Time ROS Visualization **Jan 2025 - Feb 2025**
- Developed a differential drive robot for indoor environment visualization, achieving a localization accuracy of ± 2 mm using an MPU6050 IMU and magnetic encoders.
- Designed a custom system with an ultrasonic sensor, generating a point cloud with a resolution of 1 mm per point for mapping applications.
- Implemented real-time data transmission and visualization via ROS, reducing data processing latency by 30% and enabling efficient autonomous navigation and indoor surveying.